

MACHINE LEARNING PROJECT

Notebook link:



Dataset resource:



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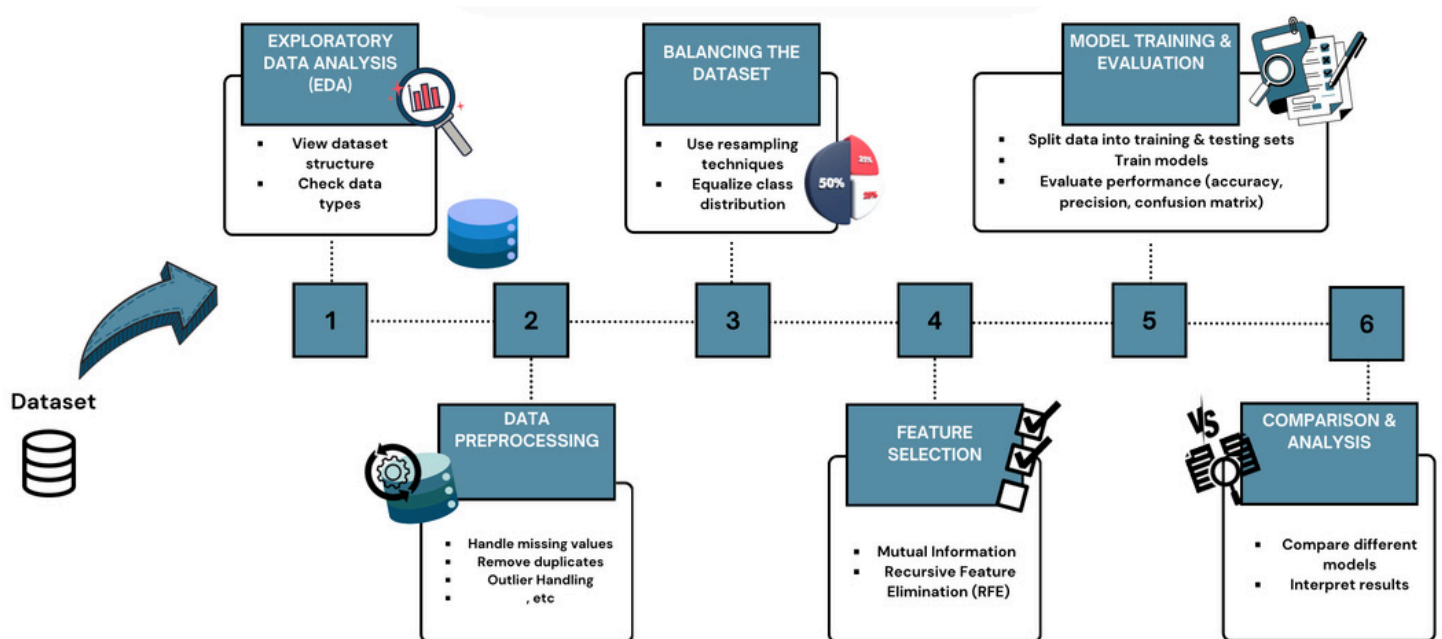
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INTRODUCTION

This report presents a comprehensive analysis of the “Predict Students’ Dropout and Academic Success” dataset, sourced from Kaggle. The study systematically covers the stages of data preprocessing, feature selection, and the evaluation of multiple machine learning algorithms. Our objective is to assess how different methodological approaches influence the predictive performance and accuracy of the models. Ultimately, we aim to identify the most effective algorithm for forecasting student dropout and academic success by leveraging a range of educational and behavioral variables. The insights derived from this analysis are intended to support academic institutions in developing proactive strategies to enhance student outcomes.

• The project workflow :



• Dataset overview:

Predict Students' Dropout and Academic dataset, is a dataset that contain The dataset contains a variety of features collected at the time of student enrollment, including academic path, demographics, and socio-economic factors . The goal of the dataset is to predict student dropout and academic success in higher education institutions..

The Predict Students' Dropout and Academic Dataset consists of **4424 instances** and **35 independent variables (features)**. The target variable are(Dropout , Graduate, and), Here's a description of the dataset:

Features:

1. **Academic path and degree program (Agronomy, Design, Nursing, etc.)**
2. **Demographics (age, gender, etc.)**
3. **Socio-economic factors (income level, family background, etc.)**
4. **Academic performance at the end of first and second semesters**

Target Variable:

- **Target:** This binary variable indicates whether the student has (Dropout, Graduate, and Enrolled).

sample of the first 5 rows:

	Marital status	Application mode	Application order	Course	Daytime/evening attendance	Previous qualification	Nacionality	Mother's qualification	Father's qualification	Mother's occupation
0	1	8	5	2	1	1	1	13	10	6
1	1	6	1	11	1	1	1	1	3	4
2	1	1	5	5	1	1	1	22	27	10
3	1	8	2	15	1	1	1	23	27	6
4	2	12	1	3	0	1	1	22	28	10

	Father's qualification	Mother's occupation	...	Curricular units 1st sem (credited)	Curricular units 2nd sem (enrolled)	Curricular units 1st sem (evaluations)	Curricular units 2nd sem (approved)	Curricular units 1st sem (grade)	Curricular units 2nd sem (without evaluations)	Unemployment rate	Inflation rate	GDP	Target
5	10	6	...	0	0	0	0	0.000000	0	10.8	1.4	1.74	Dropout
6	3	4	...	0	6	6	6	13.666667	0	13.9	-0.3	0.79	Graduate
7	27	10	...	0	6	0	0	0.000000	0	10.8	1.4	1.74	Dropout
8	27	6	...	0	6	10	5	12.400000	0	9.4	-0.8	-3.12	Graduate
9	28	10	...	0	6	6	6	13.000000	0	13.9	-0.3	0.79	Graduate

DATA EXPLORATION

Data Exploration, often referred to as Exploratory Data Analysis (EDA), is an important step in the data analysis process. It involves examining and visualizing datasets to summarize their main characteristics, often using statistical graphics and other data visualization techniques.

So, why it is important?

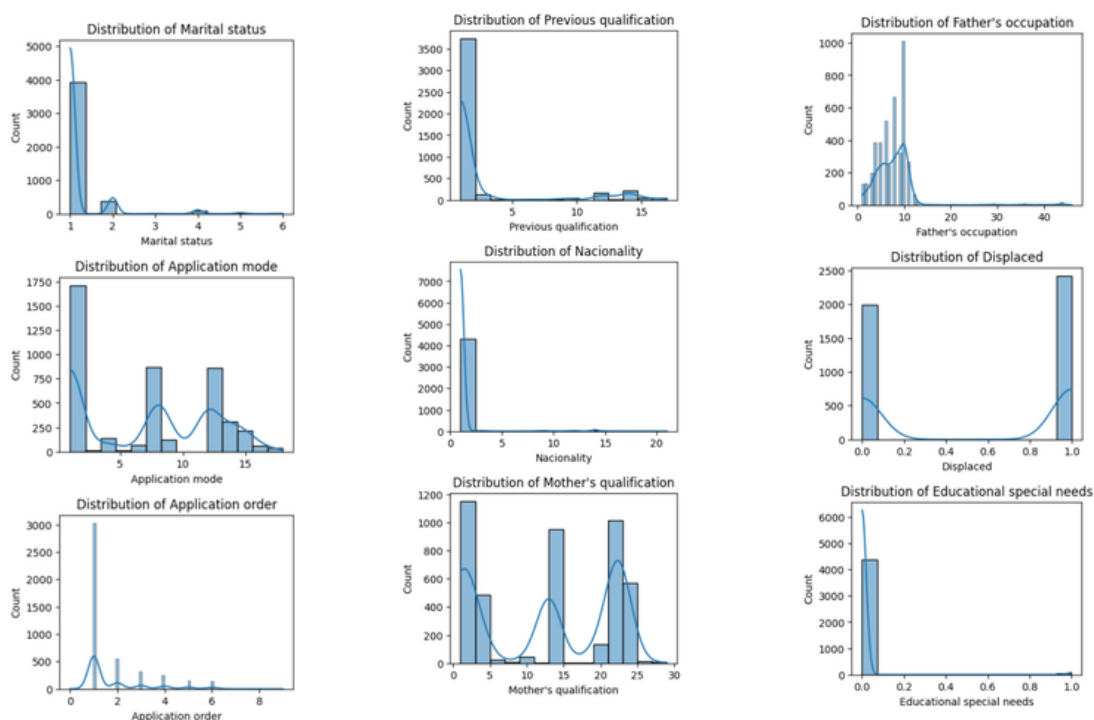
provides insight into the dataset structure and subsequent processing steps

So, Before applying machine learning algorithms and data preprocessing, an exploratory data analysis (EDA) was performed to understand the dataset's characteristics such as:

- 1- Displaying the first rows, data shape, types.
- 2- Basic Statistics: Summary statistics (mean, minimum, maximum, and standard deviation)
- 3- number of observations, features, numerical, and categorical columns.
- 5- Data visualisatio, histograms for each feature to understand the distributio.

DATA EXPLORATION

- Visualization of the data using Histogram to understand the distribution



The goal of using histograms to understand the distribution is to visually analyze how the data is spread across different values or ranges. Histograms help in identifying patterns, trends, and potential issues in the data, such as:

- Detecting Outliers: Recognizing extreme values that may require further investigation or cleaning.
- Understanding Frequency: Seeing how often certain values or ranges occur in the dataset.
- Assessing Normality: Checking if the data follows a normal distribution, which may be important for certain statistical models or algorithms.

DATA PREPROCESSING

After we performed the EDA, we did the data preprocessing, data preprocessing is a crucial step in preparing the dataset for analysis. The following steps were performed, for each step the (if-else) condition was applied to check and deal with:

1. **Handling Missing Values:** The dataset was checked for missing values, if there were missing values, we will fill them with the “mean”. But in our case there were no missing values.

2. **Removing Duplicates:** The dataset was checked for values, and confirmed that there were none. if there were duplicates, we will simply remove them. in our case there were no duplicate.

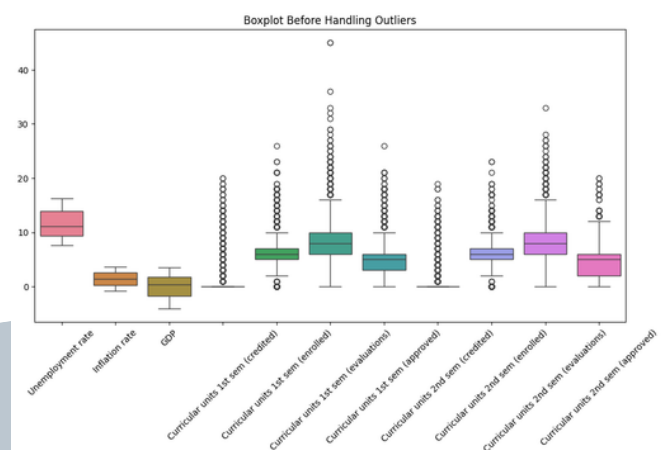
3. **Outlier Detection and Treatment:** outliers are data points that varies greatly from other results is referred to as an outlier.

Outliers were identified using the Interquartile Range (IQR) method. Features were assessed for outliers, and values beyond the lower and upper bounds were capped to mitigate their impact.

We implemented a code for outliers that counts and Boxplots the number of outliers before and after dealing with them, as follows :

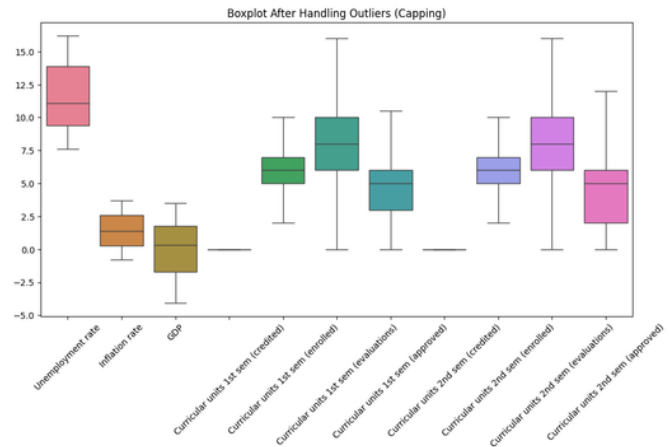
outliers before:

```
Number of outliers in each column:
Outliers in Unemployment rate: 0
Outliers in Inflation rate: 0
Outliers in GDP: 0
Outliers in Curricular units 1st sem (credited): 577
Outliers in Curricular units 1st sem (enrolled): 424
Outliers in Curricular units 1st sem (evaluations): 158
Outliers in Curricular units 1st sem (approved): 180
Outliers in Curricular units 2nd sem (credited): 530
Outliers in Curricular units 2nd sem (enrolled): 369
Outliers in Curricular units 2nd sem (evaluations): 109
Outliers in Curricular units 2nd sem (approved): 44
```



outliers After:

Number of outliers after handling (using capping):
Outliers in Unemployment rate: 0
Outliers in Inflation rate: 0
Outliers in GDP: 0
Outliers in Curricular units 1st sem (credited): 0
Outliers in Curricular units 1st sem (enrolled): 0
Outliers in Curricular units 1st sem (evaluations): 0
Outliers in Curricular units 1st sem (approved): 0
Outliers in Curricular units 2nd sem (credited): 0
Outliers in Curricular units 2nd sem (enrolled): 0
Outliers in Curricular units 2nd sem (evaluations): 0
Outliers in Curricular units 2nd sem (approved): 0



This boxplot illustrates the outliers in various features of the dataset. Below is a description of each feature based on the plot:

1. **Unemployment rate:** Values are generally clustered around the median, with a few outliers present, indicating some extreme cases.
2. **Inflation rate:** The distribution is fairly consistent, with most values close to the median and only a couple of outliers.
3. **GDP:** Shows a relatively normal distribution, but there are some outliers that indicate significant deviations from the central tendency.
4. **Curricular units 15 sem (removed):** The majority of values are concentrated around the median, with a few outliers suggesting some exceptional cases.
5. **Curricular units 15 sem (evaluation):** Similar to the previous feature, most values are around the median, with a small number of outliers.
6. **Curricular units 20 sem (received):** Values are spread out with several notable outliers, indicating a wider range of performance or engagement.
7. **Curricular units 20 sem (removed):** This feature shows a significant number of outliers, implying that some students may have extreme values.
8. **Curricular units 20 sem (evaluation):** The distribution is more uniform, with several outliers indicating a variation in evaluations across the dataset.
9. **Curricular units 20 sem (approved):** Most values are clustered closely to the median, with minimal outliers, suggesting a consistent approval rate.

4. Normalization :

- **Normalization:** means Scales the numerical features to fit them into a specific range (e.g., [0, 1]).

Min-Max scaling was applied to standardize the range of numerical features, ensuring that all features contribute equally to model training.

5. Encoding Categorical Variables: Converting categorical features into a numerical format that models can understand.

How the Code Works:

Normalization: Checks the range of numerical features. If ranges vary significantly, scaling or normalizing is required.

Encoding: Checks if there is any categorical column, if exists then One-Hot Encoding applied.

* One-Hot Encoding: Converts each category into binary columns

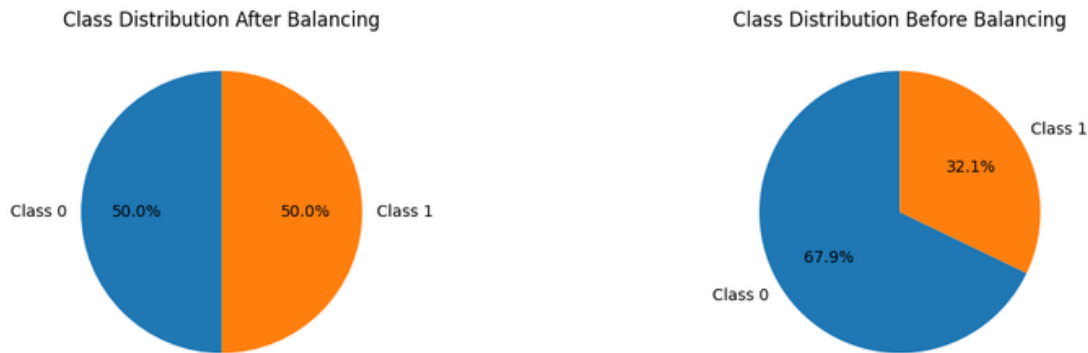
6. Handling class imbalance:

Balancing a dataset refers to the process of adjusting the number of instances of each class in a dataset

An equivalent number of samples for each class ensures that the model does not develop a bias towards any particular class

The dataset was imbalanced, meaning the number of diabetes and non-diabetes cases was unequal.

To balance the dataset, **boosting** was applied using the `resample()` function.



Boosting is an ensemble learning technique designed to improve the accuracy of machine learning models by combining multiple weak learners into a single strong learner.

How Boosting Works

1. Initialization: Start with equal weights for all training instances.

2. Iterative Training:

- Train a Weak Learner: Train a weak learner on the training data using the current weights.

- Evaluate Performance: Assess the model's performance and identify the instances it misclassified.

- Update Weights: Increase the weights of misclassified instances so that the next model will pay more attention to them. The weights of correctly classified instances may be decreased.

3. Combine Models: After a specified number of iterations (or until a stopping criterion is met), combine the predictions of all weak learners. This is often done using a weighted sum, where each model's contribution is based on its accuracy.

FEATURES SELECTION

After performing data processing, handling imbalanced a crucial step performed next, which is Feature selection is the process of identifying and selecting a **subset of relevant features** (variables, predictors) from a larger set of available features in a dataset. The primary goal of feature selection is to improve the performance of a machine learning model by enhancing model accuracy.

* **Relevance:** Selecting features that have a strong relationship with the target variable. Irrelevant features can introduce noise and make it harder for the model to learn.

Two methods were used:

1. **Mutual Information (MI):**

This method quantifies the relationship between each feature and the target variable. Features were ranked based on their mutual information scores, allowing us to select the most informative features.

- Selected top **20** features based on MI scores.

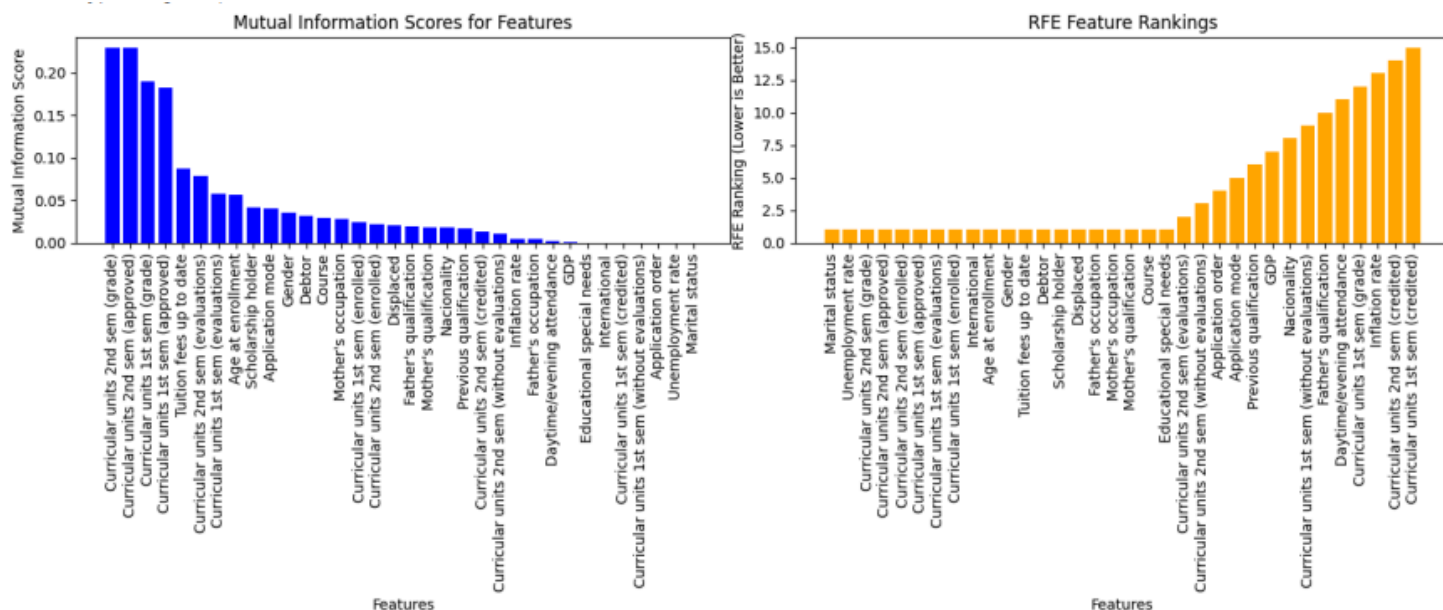
2. **Recursive Feature Elimination (RFE):**

RFE was implemented using Logistic Regression as the estimator.

This technique recursively removes the least important features based on model performance, identifying the optimal subset of features.

- Used Logistic Regression to rank and select the top 20 features

The selected features based on mutual information and RFE were then used for training the models, ensuring we focused on the most relevant data.



Performance Comparison:

Mutual Information - Accuracy: 0.85, Precision: 0.84

RFE - Accuracy: 0.85, Precision: 0.84

Mutual Information produces the best feature set.

Selected Features: ['Nacionality', "Mother's qualification", "Father's qualification", 'Displaced', 'Curricular units 2nd sem (enrolled)']

Since both methods (Mutual Information and RFE) gave similar performance with an accuracy of 0.85 and precision of 0.84, we will proceed by using the feature set selected by Mutual Information for the next step, as it provides the best feature set.

Machine learning Algorithms - 3 machine learning algorithms were implemented to predict diabetes

We trained three machine learning models on the selected features:

1. **K-Nearest Neighbors (KNN):** This algorithm classifies instances based on the majority class among the k-nearest neighbors. It is sensitive to irrelevant features and requires careful tuning of hyperparameters.

here is how we applied The knn algorithm in the code:

1- Data Preparation:

- The dataset is split into training and testing sets.
- Selected features are extracted based on prior feature selection methods.

2- Model Initialization & Finding the Best Hyperparameters:

KNN classifier initialized with a specified number of neighbors (k), set to 5.

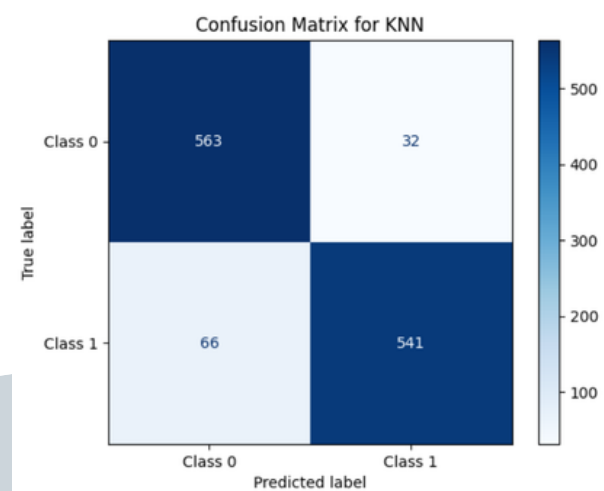
- the GridSearchCV used to find the best hyperparameters for the KNN model.
 - It explores different values of neighbors, weights (weighting of neighbors), and metric (distance calculation method).
 - GridSearchCV trains and evaluates the model with all possible combinations of these hyperparameters.
 - It selects the combination that gives the highest accuracy

3- Training & Prediction :

For a new data point, the algorithm identifies the k nearest neighbors in the training data and assigns the most common class label among those neighbors.

4- Evaluation:

The model's performance is evaluated using confusion matrix and metrics like accuracy and precision, etc.



2. Support Vector Machine (SVM): SVM finds the optimal hyperplane that separates classes in a higher-dimensional space. It is effective for both linear and non-linear classification.

here is how we applied The svm algorithm in the code:

1- Data Preparation:

Similar to KNN, the dataset is split into training and testing sets, with selected features determined from prior analysis.

2- Hyperparameter Tuning:

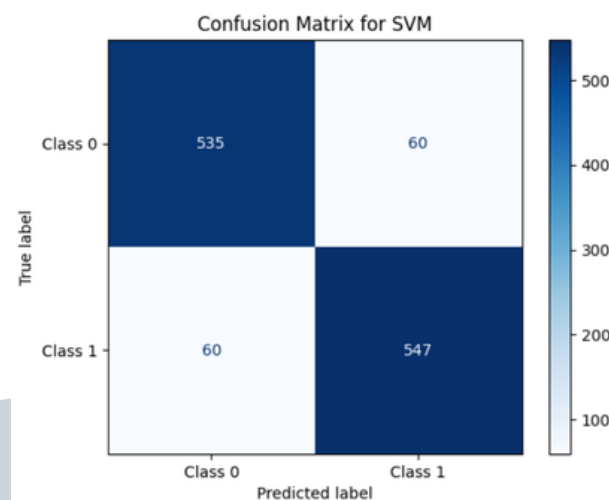
A grid search is performed to find the best hyperparameters, including the regularization parameter (C), kernel type (linear, rbf, poly), and gamma value.

3- Model Training & Prediction:

- The SVM model is trained using the training dataset with the selected hyperparameters.
- The trained model predicts the class labels for the test dataset based on the learned hyperplane.

4- Evaluation:

The model's performance is evaluated using confusion matrix and metrics like accuracy and precision, etc.



3. Decision Tree Classifier: This algorithm splits the data into subsets based on feature values, creating a tree-like model for decision-making. It is interpretable and can capture non-linear relationships.

here is how we applied The DT algorithm in the code:

1-Data Preparation:

Similar to previous algorithms, the dataset is split into training and testing sets, with selected features determined from prior analysis.

2- Model Initialization:

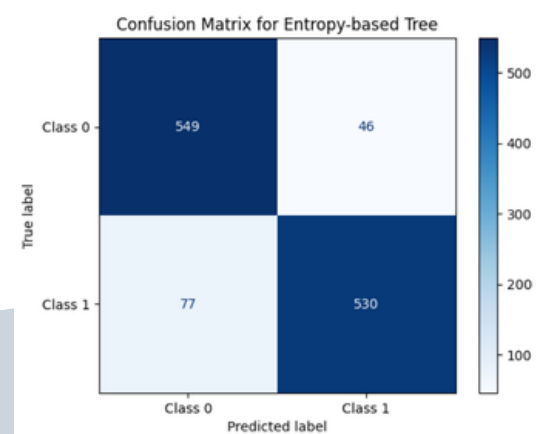
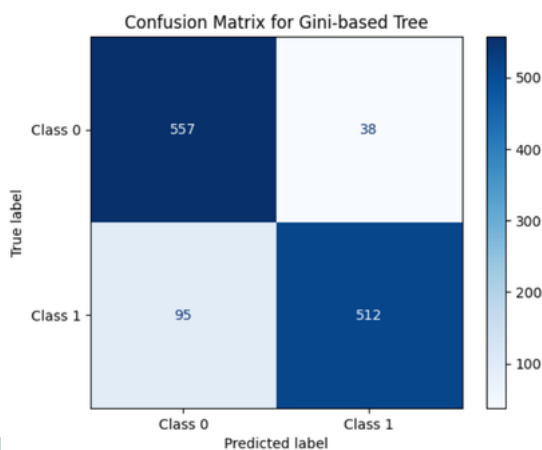
Decision Trees initialized with tow different criteria for splitting Gini impurity and entropy.

4- Model Training & Prediction:

- The model is trained on the training dataset, where it learns to make splits based on feature values to minimize impurity or maximize information gain.

- For new instances, the model traverses the tree based on feature values to arrive at a prediction.

4- Evaluation:The model's performance is evaluated using confusion matrix and metrics like accuracy and precision, etc.



MODEL COMPARISON

After training all three models, their performance is compared based on the evaluation metrics. This comparison helps determine which model performs best based on accuracy, precision, and recall.

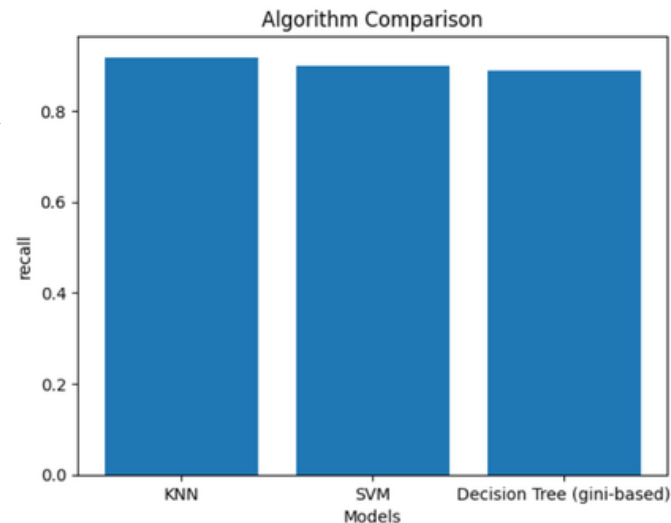
Model	Accuracy	Precision	Recall	F1-Score
KNN	0.918469	0.944154	0.918744	0.916949
SVM	0.900166	0.901153	0.900156	0.901153
Decision Tree	0.889351	0.892984	0.889351	0.889154

Each model was evaluated using a consistent set of metrics, including accuracy, precision, recall, and F1 score. The following steps were undertaken:

1. **Hyperparameter Tuning:** Grid search was employed to find the best hyperparameters for each algorithm, optimizing model performance.
2. **Cross-Validation:** A 5-fold cross-validation strategy was used to ensure robust evaluation of each model.
3. **Confusion Matrix:** The confusion matrix was computed for each model to visualize performance and understand the distribution of true positives, true negatives, false positives, and false negatives.

MODEL COMPARISON

- **KNN** achieved the highest performance among the models with an accuracy of **0.92**, precision of **0.94**, recall of **0.92**, and an F1-score of 0.9169.
- **SVM** followed with an accuracy of **0.90**, precision of **0.90**, recall of **0.90**, and an F1-score of **0.9012**.
- **Decision Tree** scored the lowest with an accuracy of **0.89**, precision of **0.89**, recall of **0.89**, and an F1-score of **0.8891**.



CONCLUSION

The KNN model provided the best accuracy for the prediction task. Feature selection improved model performance by identifying and retaining the most relevant features.

Data preprocessing (scaling, handling missing values, and encoding) was crucial for enhancing the models' predictions.

Each of these algorithms has its strengths and weaknesses, as demonstrated by the previous results. KNN provided the highest accuracy but can be computationally expensive with large datasets due to its reliance on distance calculations. SVM showed robust performance in balancing accuracy and precision but required careful tuning of hyperparameters to achieve optimal results. Decision Trees, while easy to interpret and implement, had slightly lower performance and are more prone to overfitting with complex datasets. The choice of the best algorithm depends on the dataset's specific characteristics and the goals of the analysis, such as accuracy, interpretability, or computational efficiency.